

CA-TOC — Computer Architecture (CS-328)

Weeks 1-15 · 17 Aug – 29 Nov 2026 · **Final 60** / **Sessional 40** · load 5.0 · difficulty 5/5

Read this as PDF: `python3 tools/md2pdf.py CA-TOC.md` Offsets below are line numbers into the named file.
Files live in `terms/2026-fall/CA/`. Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week.

Schedule at a glance

W	Dates	Variant	Topics
1	17–23 Aug	P2	MIPS ISA & single-cycle intro
2	24–30 Aug	P2	Single-cycle datapath
3	31 Aug–06 Sep	P2	Multi-cycle FSM control
4	07–13 Sep	P0	Microprogramming & pipeline intro
5	14–20 Sep	P0	Pipeline hazards I (structural/data)
6	21–27 Sep	P0	Pipeline hazards II (detection/branch pred)
7	28 Sep–04 Oct	P1	Midterm revision
8	05–11 Oct	MIDTERM	Exam week — no new material
9	12–18 Oct	P1	ILP & compiler techniques (recovery)
10	19–25 Oct	P1	VLIW, superscalar, SIMD/GPU
11	26 Oct–01 Nov	P0	Memory hierarchy & cache design
12	02–08 Nov	P0	Virtual memory & TLB
13	09–15 Nov	P1	RAID & advanced topics
14	16–22 Nov	P0	Final taper — past paper FIRST
15	23–29 Nov	FINAL	Exam — execute

Week detail

W1 — MIPS ISA & single-cycle intro · 17–23 Aug · P2

- **Topics:** MIPS formats (R/I/J); register conventions; single-cycle datapath design
- **Files:** narrative L17–30 · pack CA-W1 L13 · breakdown 01 (ISA, ★★★★★)
- **Drill:** decode 10 instructions to binary; draw single-cycle datapath <15 min (Diag #1)
- **Trap:** `lw` is the critical path — all instructions share its cycle time; I-type sign extension is per-format

W2 — Single-cycle datapath · 24–30 Aug · P2

- **Topics:** multi-cycle vs single-cycle; multi-cycle datapath for R-type/lw/sw/branch
- **Files:** narrative L33–46 · pack CA-W2 L21 · breakdown 02 (single-cycle, ★★★★★)
- **Drill:** trace lw/sw/beq/R-type step-by-step; draw multi-cycle datapath <15 min (Diag #2)
- **Trap:** missing wait state = lost marks; IR/MDR/A/B/ALUOut — which needs a write control signal?

W3 — Multi-cycle FSM control · 31 Aug–06 Sep · P2

- **Topics:** FSM for multi-cycle control; state transitions; ASM charts
- **Files:** narrative L49–64 · pack CA-W3 L28 · breakdown 03 (multi-cycle, ★★★★★☆)
- **Drill:** FSM state diagram for R-type/lw/sw/branch, every transition labeled (Diag #6)
- **Trap:** every instruction must return to state 0 (instruction fetch); missing transition = processor stalls

W4 — Microprogramming & pipeline intro · 07–13 Sep · P0

- **Topics:** horizontal vs vertical microprogramming; 5-stage pipeline; pipeline registers
- **Files:** narrative L68–81 · pack CA-W4 L35 · breakdown 01/02
- **Drill:** 5-stage pipeline, trace 3 instructions (Diag #3); microprogramming comparison table
- **Trap:** ideal 5-stage speedup = 5 only with zero hazards; structural hazard = one shared resource

W5 — Pipeline hazards I · 14–20 Sep · P0

- **Topics:** structural hazards; RAW/WAW/WAR; forwarding unit design
- **Files:** narrative L85–100 · pack CA-W5 L42 · breakdown 04 (pipeline-hazards, ★★★★★)
- **Drill:** identify hazards in 8 sequences; design forwarding unit (Num #13–14)
- **Trap:** RAR is NOT a hazard; WAR = read-before-write; WAW = two writes; `$zero` must never be forwarded

W6 — Pipeline hazards II · 21–27 Sep · P0

- **Topics:** hazard detection unit; stall vs forward; 1-bit/2-bit branch prediction; control hazards
- **Files:** narrative L104–117 · pack CA-W6 L51 · breakdown 04
- **Drill:** 2-bit predictor trace, 10 branch outcomes (Diag #7–8); hazard detection stall decisions
- **Trap:** 1-bit predictor mispredicts twice per loop (first + last iteration); 2-bit only once

W7 — Midterm revision · 28 Sep–04 Oct · P1

- **Topics:** comprehensive review of weeks 1–6
- **Files:** narrative L121–137 · pack CA-W7 L60 · breakdowns 01–04
- **Drill:** timed past paper 2 hrs; multi-cycle draw 12 min blank; FSM design; 3 hazard sequences
- **Trap:** no re-reading — blank page first; if multi-cycle >15 min, repeat 5× daily until exam

W8 — MIDTERM EXAM WEEK · 05–11 Oct

- **Focus:** active recall only. No new material.
- **Files:** narrative L141–149 · no pack (L68) · ledger frozen
- **Stack:** past paper 60 min → blank page 30 min → error log 20 min
- **Trap:** answer diagram/numerical first, theory second; check every control signal

W9 — ILP & compiler techniques · 12–18 Oct · P1 (recovery)

- **Topics:** ILP; Amdahl's Law; ILP/power/memory walls; loop unrolling; software pipelining
- **Files:** narrative L152–171 · pack CA-W9 L71
- **Drill:** unroll loop 4× closed-book; software pipeline prologue/kernel/epilogue
- **Trap:** Amdahl — speedup = $1/((1-f)+f/n)$; even infinite processors cap at $1/f$

W10 — VLIW, superscalar, SIMD/GPU · 19–25 Oct · P1

- **Topics:** VLIW; superscalar; in-order vs out-of-order; SIMD vs SIMT; GPU architecture
- **Files:** narrative L175–189 · pack CA-W10 L79
- **Drill:** VLIW packet scheduling; superscalar IPC; SIMD stride; warp divergence utilization
- **Trap:** IPC can never reach issue width — branches and dependencies bound it

W11 — Memory hierarchy & cache design · 26 Oct–01 Nov · P0

- **Topics:** direct/set-associative/full-associative; write policies; AMAT
- **Files:** narrative L193–207 · pack CA-W11 L88 · breakdown 05 (cache-VM, ★★★★★☆)
- **Drill:** 10 cache numericals — address breakdown, miss rate, AMAT (Num #5–8)
- **Trap:** compute tag/index/offset split first; stride vs sequential access; write-back vs write-through traffic

W12 — Virtual memory & TLB · 02–08 Nov · P0

- **Topics:** virtual memory; page tables; TLB; EAT; multi-level paging
- **Files:** narrative L211–228 · pack CA-W12 L95 · breakdown 05³

- **Drill:** page table size; TLB reach; EAT (Num #9–12); two-level page walk
- **Cross-course:** TLB ↔ OS (CS-329) — same concepts, check your OS notes here
- **Trap:** EAT is dominated by page-fault time (5 ms vs 100-cycle memory); TLB reach = entries × page size

W13 — RAID & advanced topics · 09–15 Nov · P1

- **Topics:** RAID 0/1/4/5/6/10; RAID performance; I/O systems
- **Files:** narrative L232–246 · pack CA-W13 L102
- **Drill:** RAID comparison table from memory (Diag #13); RAID 4 vs 5 write penalty
- **Trap:** RAID 4's dedicated parity disk is the bottleneck RAID 5 fixes; write penalty analysis

W14 — Final taper · 16–22 Nov · P0

- **Topics:** cumulative review of weeks 1–13
- **Files:** narrative L250–269 · pack CA-W14 L109
- **Drill:** full past paper 3 hrs — FIRST, not after review; 10 hazard timing diagrams; numerical speed drills
- **Trap:** no passive reading all week; every minute is active retrieval; sleep 9 hrs, never trade sleep

W15 — FINAL EXAM · 23–29 Nov

- **Focus:** execution. Learn nothing new.
- **Files:** narrative L273–283 · no pack (L118) · cheat sheet from W14
- **Stack:** numericals → diagrams → theory; label every forwarding path
- **Trap:** flag stuck problems and move on; return if time permits

Deep-study index

Book	What it holds	Open with
Diagram-Book.md	20 diagrams; canonicals #1–15	Draw from blank until <10 min
Numerical-Book.md	25 problems (#1–15 core, #16–25 extended)	Closed-book retrieval, formulas first
Formula-Book.md	Formulas grouped by topic	Re-derive from first principles
Definition-Book.md	Definitions + short theory answers	Recall aloud after each week
Top-100-Questions.md	100 Qs in 5 buckets	Buckets: ISA/Datapath L3–29, Pipelining L31–61, Cache L63–84, Advanced L86–103, Theory L105–115
Top-10-Tricky-Concepts.md	10 highest-yield pitfalls	Before each past paper
03-Chapter-Breakdowns/	5 chapters with weight stars	01/02/04 ★★★★★ · 03/05 ★★★★★☆